

Automated Performance-Scoring Platform

Replacing a manual, screenshot-based leaderboard with a governed, warehouse-driven scoring system and a live display

Role: Data and automation lead (scoring layer, data quality, dashboard integration) **Type:** Internal analytics product

Duration: About 1 week, build to launch

1. Overview

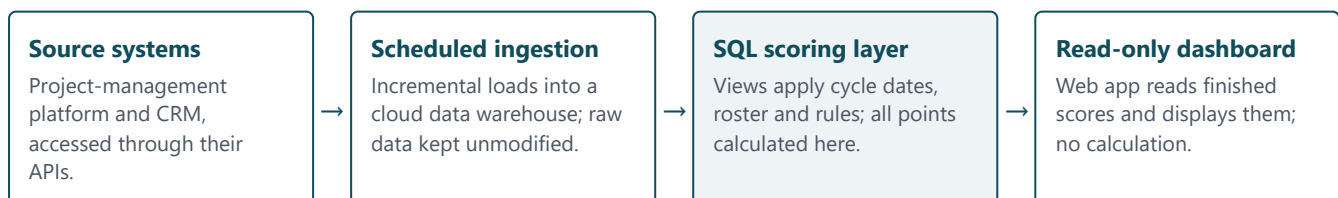
An organisation ran a recurring, gamified performance leaderboard that tracked activity in two systems: a project-management platform and a CRM. Scores were produced by hand every fortnight. I designed and delivered a replacement: automated ingestion into a cloud data warehouse, a configuration-driven SQL scoring layer, and a read-only web dashboard that refreshes automatically on a large office display.

Manual → automated Fortnightly copy and paste replaced by scheduled refresh	Twice daily Data refresh cadence, display updates within minutes	17 SQL views Scoring, configuration, freshness and data-quality layers	100% reconciled Displayed totals matched the warehouse at launch
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2. The problem

- **Manual and slow:** screenshots of dashboards were converted to figures and pasted into an admin panel.
- **Inconsistent windows:** the source dashboards could not be set to the cycle dates, so numbers rarely matched the period being scored.
- **Opaque logic:** points were calculated inside the display application by an AI agent, so nobody could verify how a score was reached.
- **Two different kinds of activity** (task activity and sales pipeline stages) were being combined without clear rules.

3. Solution architecture



Ingestion was owned by a data engineer; I partnered on requirements and owned everything from the scoring layer onward.

4. Key design decisions

Decision	Reasoning
Calculate in the warehouse, display elsewhere	One auditable source of truth. The application became display-only, which removed a whole class of silent calculation errors.
Rules as data, not code	Point weights, scored pipelines, cycle anchor and roster live in configuration views, so rule changes take minutes and need no redeployment.
Self-rolling cycles	A single anchor date and fixed length generate every cycle with no gaps or overlaps; the display rolls over automatically.
"Entered stage and still in stage" counting	Pipeline events are counted by unique deal identifier, which prevents double counting and keeps results explainable.
Time zone discipline	All cycle boundaries are evaluated in the business time zone, not UTC, so late-evening activity lands in the right cycle.
Region constraint handled, not ignored	The two datasets sat in different warehouse regions and could not be joined. Each source was scored separately, and the only combination (an individual total) is a documented addition in the display layer.

5. Security and governance

- **Least privilege:** the dashboard uses a dedicated read-only service account that can query only the published scoring views.
- **Authorised views:** scoring views read the raw tables on the account's behalf, so the dashboard can never see raw records.
- **Controlled write access:** the raw tables were never altered. Roster enrichment used additive, clearly prefixed columns agreed with the data owner, with documented rules so the ingestion job could not be broken by them.
- **Secrets hygiene:** credentials were stored only in the hosting platform's secret store and never handled in chat or documents.

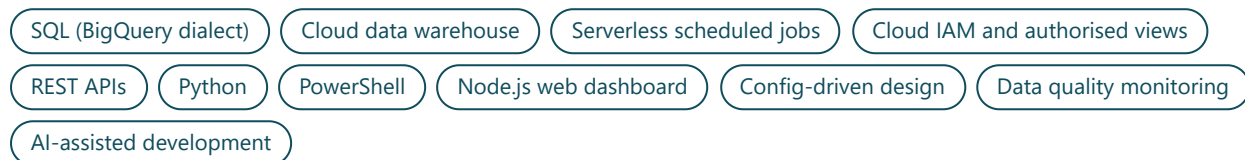
6. Data quality and operations

Control	Purpose
Freshness views	Each feed reports OK, WARN or STALE from the age of its last successful load, tuned to the refresh schedule.
Unscored-record report	Lists every record excluded from scoring and the reason, so gaps are visible rather than silent.
Unclassified-user report	Flags active users with no recorded team or role the moment they appear.
Independent recomputation	Team totals were rebuilt from the raw tables without using the scoring views and had to tie out exactly before launch.
Diagnostics toolkit	A reusable PowerShell and SQL toolkit runs a full read-only health check of both feeds, scores and access in one command.
Resilient display	If a refresh fails, the display keeps the last good scores instead of showing blanks or zeros.

7. Outcomes

- Replaced a fortnightly manual process with an automated, twice-daily refresh.
- Made every score traceable to the individual task, comment or deal that earned it.
- Surfaced and fixed data issues before launch: duplicate-looking records, undated pipeline stages, username mismatches between systems, and a stalled feed that was detected by the freshness controls.
- Delivered a clean office-display experience: fit-to-screen layout, and automatic scrolling for long lists.
- Handed over documentation, a to-do register and a reusable skill so the system can be maintained by others.

8. Technologies



9. Skills demonstrated

- Analytics engineering: modular, layered SQL with configuration separated from logic.
- Data governance: least privilege, auditability, controlled change management.
- Stakeholder management: turning loosely stated rules into precise, signed-off definitions across sponsors, data owners and end users.
- Operational thinking: monitoring, runbooks, rollback paths and handover documentation.

10. CV-ready summary lines

- Designed and delivered an automated performance-scoring platform, replacing a manual fortnightly process with a governed warehouse-driven pipeline and a live office dashboard.
- Built a configuration-driven SQL scoring layer (17 views) combining project-management and CRM activity under auditable rules, with self-rolling cycles and time-zone-correct reporting.
- Implemented least-privilege access (read-only service account, authorised views) and data-quality controls (freshness monitoring, unscored-record and unclassified-user reports).
- Verified results through independent recomputation and launch-day reconciliation; produced runbooks, diagnostics tooling and handover documentation.

All names, figures and systems in this document are generalised. It describes an illustrative architecture and contains no organisational data.